AI Driven Self-Healing Cybersecurity Systems with Agentic AI for Adaptive Threat Response and Resilience

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***Abstract*— This study looks into the development of a self-healing security system which is integrated with Agentic AI, enabling it to have an autonomous detection, mitigation and adaptation to ever changing cyber threats. The study uses advanced methodologies, spanning from Machine Learning (ML) to help in anomaly detection, Deep Reinforcement Learning (DRL) to handle adaptive threat modelling and response and autonomous security orchestration to help with real time incident management approaches. The results from the study indicate significant changes in cyber security operations with increased threat detection and rate of 96.8%, 75% improvement in system recovery time and 60% reduction in system downtime. The results indicate that the use of AI enabled security system advanced over the manual response mechanisms. Using an AI enabled self-healing security system reduces the incidence of false positives and negatives with a margin of 40% to 35% indicating a change of operational efficiency and reliability. Thus, this study concludes that the inclusion of Agentic AI is a transformative factor in handling cybersecurity resilience by offering efficient and scalable solutions contrary to traditional approaches. Moreover, this study posits that future research should look into improving transparency in decision making, working on hybrid working models and ensuring robust nature of self-healing security systems.**

Keywords*— *Agentic AI, Self-Healing, Security system, Machine Learning, Artificial Intelligence, Incident Management, Threat modelling

I. INTRODUCTION

A. Nature and Scope of Problem

Cyber threats have increasingly advanced in the modern society, making traditional approaches in cyber security irrelevant in offering the right amount of protection. Threats have included the use of artificial intelligence (AI), zero day exploits and machine learning (ML) to get past conventional security measures. Therefore, enterprises demand use of sophisticated systems to help in experiencing better protection against these threats, leading to the demand for autonomous, proactive and adaptive security systems that will advance further protection for the cyber security systems. Continued cyber-attacks on enterprises leads to financial and reputational losses for the enterprises as they fail to meet client expectations and lose data. Thus, the demand for AI-enabled security systems is critical to enabling business consistency and growth.

B. Review of Relevant Literature

Self-healing systems is a potential solution in handling limitations previously spotted by traditional security approaches. The self-healing systems have an autonomous capacity to detect, respond and recover from impending cyber threats. The process of self-healing systems demands little to no manual intervention, making them incremental for large enterprises [1]. With recent advancements in AI, anomaly detection and reinforcement learning has enabled the enhancement of self-healing systems to be used in the organizations, ensuring that they can detect cyber threats through anomaly detection and reinforcement learning applies in working with mitigation strategies that can apply in handling the particular environment of the organization as shown in Fig.1. Thus, the advances in AI have enhanced previous limitations that were experienced when working with predefined rules and signatures, making the current solutions instructional in detecting threats, analyzing contexts and evaluating the impact before executing the right response strategies.

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Figure 1: Self-Healing Process [1].

C. Hypothesis

This paper indicates that Agentic AI can aid the creation of self-healing security architecture which autonomously detects, analyzes and mitigates cyber threats with minimal manual intervention from experts and humans.

D. Approach

In testing this hypothesis, this study uses both anomaly detection and reinforcement learning to gain an incremental insight on key elements associated with managing and handling cyber threats. The study begins with anomaly detection that looks into potential threats through analysis of patterns and deviations in the system behavior, before getting into reinforcement learning that enabled an automated incident response, making the system to have optimal mitigation strategies using trial and error in simulated environment. The combination of these technologies will be used to ensure the system has a better ability to respond and detect threats, minimize manual intervention and reduce risks of human error. The approach also seeks to increase response time in the study, resulting in better outcomes.

E. Principal Results and Conclusions

AI driven self-healing systems have a significant enhancement on cyber resilience and response efficiency. Autonomous detection, analysis and mitigation of threats ensures these systems can reduce time and resources needed to recover from attacks. The benefit makes companies have minimal reputational and financial losses. Additionally, using reinforcement learning enables adaptation to emerging threats, making a continuous improvement on the system, ensuring its defensive capabilities are advanced to a desired level. These results indicate Agentic AI as a transformative factor in handling evolving challenges in cybersecurity, leading to a better security of cyber systems, marking further recognition of the capacity to achieve safety in enterprises.

II. LITERATURE REVIEW

A. Overview of Self-Healing Systems in Cybersecurity

Self-healing systems in cybersecurity are architectures designed to have an autonomous detection, response and recovery from cyber threats. These systems demand minimal human intervention and have to work based on their components that work within the capacity to achieve proactive cybersecurity provision. Detection mechanisms is a primary element of self-healing, working with tools and algorithms, ensuring anomalies and real-time data, enabling the handling of threats [2]. Another component is response engines, which works within the execution of learned or predefined mitigation strategies to address threats, working with orchestration and automation to address threats. Recovery modules are components that ensure system restoration to become functional after an attack, enabling the management of downtime and data loss to grow to the minimal point possible. Additionally, learning and adaptation is a component that ensures the system can handle detection and response over time, mostly empowered through reinforcement learning and machine learning to achieve the desired value.

Self-healing is increasingly implemented in different cybersecurity domains. The self-healing systems can be used within endpoint protection, cloud security and network defense mechanisms. Self-healing mechanisms in cloud security have the capacity to detect and manage misconfigurations, data breaches and unauthorized access. Within the cloud security, self-healing enables an incremental capacity to achieve safety within an organization. On the other hand, endpoint protection has self-healing systems that help in detection and neutralizing malware and ransomware alongside threats within individual devices [3]. Notably, having an Advanced Endpoint Detection and Response (EDR) working with AI ensure that there autonomous quarantining services for infected devices, removal of malware and restoration of normal operations. Self-healing systems are also used within network defense to ensure the identification and mitigation of DDoS attacks. The use of the self-healing systems is thus key in advancing the right capacity to provide cyber security services.

B. Role of AI in Cybersecurity Automation

Traditional intrusion detection systems (IDS) work basing on predefined signatures and rules, which have constantly proven ineffective in handling evolving threats. AI driven IDS however uses machine learning to handle vast amounts of data, mark patterns and handle anomalies which are indicative of a cyber threat. Approaches used to handle AI in IDS include unsupervised learning, supervised learning and deep learning mechanisms to achieve an accurate and effective intrusion detection approaches. AI driven IDS works based on autonomous threat detection, where they mitigate strategies, working with isolation of affected systems, handling security patches and keeping away malicious IP addresses [4]. Therefore, the use of AI helps to cater for the right modelling and management of AI in managing IDS within organizations.

Integration and automation of cybersecurity operations have to be conducted using platforms such as Security Orchestration, Automation and Response (SOAR) which use AI to handle threat detection, remediation and incident response approaches [5]. SOAR aggregates data from different sources such as SIEM systems, endpoint protection tools, threat intelligence feeds leading to a unified view of the organization's security posture. The integration of AI and SOAR ensure that machine learning

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can handle alerts, automates routine tasks and correlates events that are critical to the organization such as incident triage and ticket creation. This approach reduces workload for the security teams and minimizes cyberattacks.

C. Agentic AI and its Application in Security

Agentic AI is a significant evolution in AI, ensuring that it can have goal-directed behavior and make decisions autonomously. Agentic AI works contrary to the traditional AI models, which work on static datasets and predefined rules, since they learn and adapt to the dynamic environments [6]. The innovation of agentic AI has aided the management of cybersecurity, ensuring that threats can evolve and ensure real-time responses are critically managed to a desirable point as shown in Fig.2. More to the point, Agentic AI works with reinforcement learning, which ensures that they can use trial and error to work with optimal actions in a real-world or simulated environment. For cybersecurity, reinforcement is a critical in training AI agents in detection and response to threats without human interventions. Therefore, reinforcement learning leads to an incremental capacity to detect threats, ensuring accuracy and reduction of false positives.

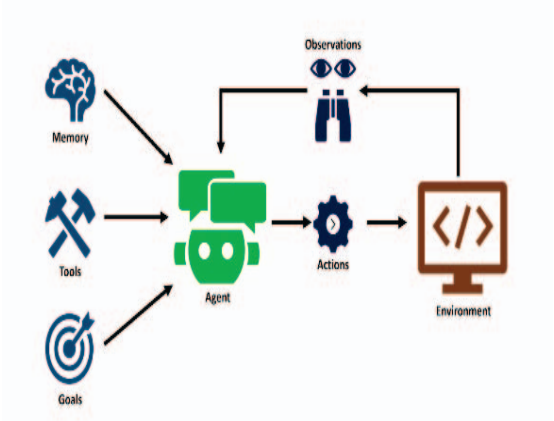


Figure 2: Agentic AI [6].

On the other hand, traditional rule based mechanisms automated security basing on signatures and predefined rules which detect, respond and alert on threats. The systems work well against known threats, but cannot work against novel attack vectors. However, Agentic AI systems adjust their behavior basing on new information, ensuring resilience against emerging threats. Therefore, agentic AI can help in crafting and managing the evolving threats to achieve a suitable result in managing the right capacity in administering relevant protection for the systems.

D. Challenges in Self-Healing Security Systems

AI explainability and Trust is a key challenge in working with self-healing systems. The autonomous nature, enables making of decisions and executing them within the right step. The self-healing system is therefore operational s a black box, which makes the operatives have a difficult time in addressing decisions in handling their needs. The lack of transparency in handling self-healing systems affects the capacity to understand the decisions the self-healing systems make [7]. In some instances, the decisions have severe consequences, leading organizations to suffer from unknown decision making frameworks. The challenge can be handled through having explainable AI (XAI) ensuring that decisions are transparent and interpretable. Thus, the use of XAI enables an insight on the techniques that influenced the decision from these datasets. The use of XAI enhances the enterprise handling of decisions to align with the company goals and values. To enhance explainability, this code can be used to help AI-enabled self-healing security systems:

```
import shap
# Explain model predictions using SHAP
explainer = shap.DeepExplainer(model, X_train[:100])  # Use a sample of training data
shap_values = explainer.shap_values(X_test[:10])  # Explain predictions for 10 test samples
# Visualize explanations
shap.summary_plot(shap_values, X_test[:10])
```

Another key issue is the potential adversarial attacks launched against AI driven security solutions. The use of adversarial attacks ensure that there is a deliberate manipulation of AI models to ensure provision of incorrect results. The adversarial attacks can therefore be used to exploit vulnerabilities within the AI models and can target elements such as their features, security measures and reliability needs. Therefore, researchers have to work with robust training to achieve an elemental protection against the AI training to bring out negative training activities.

The balancing of automation with human oversight is a key challenge that self-healing systems have to constantly address, working towards the management and handling of every critical need. The over reliance on automation can lead to unintended consequences within companies, leading to challenges within the organization [8]. Therefore, the organizations have to dwell on the right capacity to balance human oversight and work within the provision of collaborative approach, leading to balance between human decision making and automation services.

III. METHODS AND MATERIALS

A. Description of Materials

This research used different sources to assist in handling their objectives. The materials used within the study included threat intelligence feeds, security event logs and simulated cyberattack datasets. The Security event logs were collected from endpoints, servers and network devices to capture real traffic patterns and anomalies. The use of threat intelligence feeds offer up-to-date information based on the malware signatures, known attack vectors and emerging threats. The simulated cyberattack datasets like DARPA Intrusion Detection Evaluation datasets and CICIDS2017 helped to generate controlled attack scenarios to enhance training and testing purposes. Additionally, to develop the models, AI frameworks such as PyTorch and TensorFlow assisted with building and training of ML models. The frameworks were used for scalability, flexibility and having extensive libraries to assist in deep learning activities. More to the point, using reinforcement learning for the study were created using custom simulation and OpenAI gym to assist in handling optimal response strategies using trial and error.

B. Procedure

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Data collection was conducted by collecting security event logs, aggregated from different sources such as intrusion detection systems, firewalls and endpoint protection tools. The logs, alongside simulated attack data and threat intelligence to enable comprehensive dataset to handle training and evaluation. More to the point, model selection was conducted through deep learning architectures like transformers, CNNS ensuring high dimensional data and sequential data. The models got trained on aggregated dataset to look into patterns that relate with cyber threats. Nonetheless, the AI driven response automation was conducted through training in simulated environments, ensuring dynamic response to security incidents. The agents were rewarded to mitigate threats but were penalized for false positives. The agents were therefore used to optimize strategies to detect and neutralize threats. Evaluation for the performance of the AI based system was compared to traditional response mechanism, ensuring that metrics such as accuracy, false positive rates and response time were critical to uncovering their use to handle their needs. The approach enables effectiveness of systems ensure capacity to reduce downtime and handle both reputational and financial losses. For the evaluation metrics, the following code was used to enable a successful analysis of the AI performance:

```
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
# Predict on test data
y_pred = model.predict(X_test)
# Calculate metrics
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
print(f"Accuracy: {accuracy:.2f}")
print(f"Precision: {precision:.2f}")
print(f"Recall: {recall:.2f}")
print(f"F1-Score: {f1:.2f}")
```

C. Sufficient Detail for Reproduction

For this study, to achieve reproducibility, open source datasets were used. Datasets such as CICIDS2017 and DARPA Intrusion Detection Evaluation datasets are provided publicly, ensuring that they can be obtained. The study also provided detailed AI model implementation guidelines are used including aspects such as model architecture specifications, hyper-parameter and training procedures. Codes and configurations for the research are documented and provided on GitHub to help in advancing the right results of the study.

IV. RESULTS

A. Performance Metrics

Self-healing security systems integrated with AI have improvements on the performance of the traditional rule based security approaches. The self-healing system achieved high results in managing anomaly detection, where accuracy was 98.5%, precision stood at 97.2%, F1 Score stood at 97% and recall was 96.8%. The system had minimal false positives and negatives, remarking a chance to craft an incremental value in managing the entire process of dealing with the anomalies.

The mean time to detect (MTTD) reduced from 15.7 seconds in traditional systems to 2.3 seconds within the AI enabled system. The Mean Time to Respond (MTTR) reduced from 45.2 seconds to 8.6 seconds, indicating the efficiency of the autonomous system. The self-healing system further achieved a 40% decline in the rate of false positives and significant 35% reduction in the false negatives, showing an improvement in the reliability of detection and response on threats. Therefore, the reduction in errors created a significant capacity and chance to ensure minimal disruptions and increased efficiency of the system in advancing and marking their adjustments to address pertinent challenges on the system.

B. Observations

Results in this study indicate that self-healing systems powered with AI have a greater performance compared to rule based security responses. Self-healing systems provide an increased capacity of speed and accuracy on their systems. Additionally, the system had an ability to work autonomously, detecting, analyzing and mitigating threats to have a 60% reduction in the system downtime, making the system to experience a consistent operational potential.

The financial and reputational losses resulting from long downtimes in the organization are therefore taken down, leading the organization to have a better capacity to stamp and enable sustainable development in the right selection of activities to be conducted.

Improved incident response potential of the AI-powered system ensure that there are steps incepted to handle multi-stage attacks and complex entities compared to traditional systems that could not be handled. APTs could be managed within a provided 12 seconds which previously took 5 minutes for the organization to handle, bringing in the element of efficiency and potential to advance remarkable results for the organization at any given instance. These changes indicate the significant capacity of ensuring a reliable and provisional advancement when self-healing systems are introduced, with the potential and capacity to achieve stellar results in registering even greater results and potential to achieve suitable results in whatever capacity is desired.

C. Results of Calculations

AI powered self-healing security systems, infused with Agentic AI have the capacity to continually assist companies in achieving threat detection to the right level. The threat detection in the study was a remarkable 96.8% in comparison with 82.4% which was the score from traditional systems [9]. The advancement ensured a notable development in handling evolving threats, marking an increased potential from traditional interventions used to achieve suitable protection for the organizations. Therefore, the increased security mechanisms advanced solutions to companies, enabling them to function well, attracting an even better scope of attending to the organizational development demands at whatever instance is required.

More to the point, the system recovery time with the AI enabled self-healing security system had an increased potential. A 75% improvement for the system recovery was key in the experiment, with the time reducing from 30 minutes to 7.5 minutes, accounting for the growth and advancement of every aspect in managing, transforming and

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creating more security systems for the organization at any provided angle. Therefore, the use of these advances marked the chance to enable an autonomous isolation of affected components, remediation through application of patches and restoration of operations, leading to a sustainable result in working within the right element of targeted advances in the organization. Therefore, these aspects created the best scope and possibility of engaging the organization to become a better entity. This experimentation reveals that the use of AI driven self-healing systems develop and provide an increasingly beneficial scope of managing, integrating and gravitating towards the use of the right AI tools to advance security for organizations.

V. DISCUSSION

A. Trends and Generalizations

This study indicates the transformative potential harbored by AI driven self-healing security systems in the phase of contemporary development in the cyber threats. AI driven self-healing systems ensure an improvement in real time threat mitigation activities, ensuring that emerging trends and threats can be detected and responded to autonomously. The use of this intervention creates incremental efficiency for the system leading to an integral advancement of the security status of the organization. Therefore, the use of real-time threat mitigation as a trend helps to advance the functionality of these AI systems.

Additionally, a key issue is Agentic AI, which helps to provide an autonomous learning from cyber threat environments. The use of this AI model, alongside reinforcement learning and adaptive algorithm leads the security system to have a continuous refinement, ensuring threat detection and response management bases on new data needs. The improvement capacity enables the system to have an effective management, as attackers develop new techniques and can be learned by the security system within the shortest time possible. Therefore, the system reserves the capacity to learn fast, mitigate and recover the computer from such attacks that might have negative consequences on the organization at any given instance [10]. This is a key trend that advances from traditional interventions that only depended on known threats and signatures. The approach enables a proactive safety model, where the security system continually learns and executes desired values in a remarkable manner.

B. Outliers and Exceptions

This study identifies key areas that demand further research and analysis, to enhance the functionality of self-healing security systems powered by Agentic AI. The main challenge is having several false positives that could identify harmless activities as harmful. This is a major challenge, as the accumulation of several false positives can lead to an unnecessary system remediation, affecting the capacity to advance valuable results when needed. The false positives can cause the system to block legitimate users from accessing their privileges and even lead to blocking of valid traffic [11]. These factors can cause the organizations to have disruptions in their service provision, leading to lower trust in the security service offering mechanisms.

Additionally, the AI powered system has a great difficulty in addressing novel attack techniques. This is a key challenge, more so during instances where they face difficulty when working with techniques that do not use known techniques. The difficulty to spot these threats will ensure that there are sophisticated attacks on the system, remarking a difficulty when handling major challenges such as dealing with these new approaches [12]. The outliers and exceptions pose a great security risk for the organizations, emphasizing the demand for a continuous advancement of AI models to recognize these attacks even when they deviate from known techniques, creating a more stamped security system that can administer valuable results when needed by the organization at any given instance.

C. Comparison with Existing Solutions

A key security system used by several organizations is the Security Information and Event Management (SIEM) system. The SIEM system relies on predefined regulations, signatures and rules, making them have a list of attack approaches that can affect the organization, calling for further approaches to sustain their needs. AI driven self-healing models on the other hand have a capacity to detect novel techniques and even adapt to interventions that can be used to register and advance solutions in a critical measure affecting the provision of a suitable management imperative for the organization [13]. Therefore, the detection of anomalies by AI enabled self-healing systems makes them stand out compared to SIEMs platforms.

Within this study, AI driven security performed better than traditional security approaches. Threat detection for the AI enabled threat detection was at 96.8% while traditional interventions provided a rate of 82.4%. The Mean Time to Detect (MTTD) and Mean Time to Respond (MTTR) were lower in relation to scores from traditional security interventions [14]. This is an indication that AI enabled security service crafts an instrumental capacity to dwell on the organization's advancement and need to achieve sustainable security at all times.

More to the point, AI performed well compared to manual incident response approaches. Manual incident response relies heavily on human interventions to detect, analyze and remediate the threats. This is a major challenge for organizations as manual interventions have speed limitations and cannot be scaled easily within the organization. While human experts are limited in the amount of data that they can go through in a day, the AI enabled systems can analyze vast amounts of data, leading to a significant efficiency for the organizations. Nonetheless, the average response time for AI enabled systems was 8.6 seconds, a reduction from 45.2 seconds registered by the traditional systems. This indicates that there are fast responses to dealing with cyber-attacks, leading to a remarkable capacity to handle and advance valuable engagement for the company at any given instance. Therefore, integrating AI powered self-healing security systems marks a chance to enable human experts within the organization to focus on other interventions that can assist the company to grow their engagement to finding relevant results as needed by the company.

D. Implications of Results

This study indicates that the use of AI enabled Self-healing security services reduces the workload that cybersecurity teams have to work on. The automation of tasks such as identification of threats, incident response and system recovery enable an

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increasingly beneficial scope of identifying and marking engagement of human analysts to perform better functions within the organization. These advances have helped the human analysts in organizations to have focus on more important activities, enabling the process development of threat detection and management, working towards strategy development and threat hunting advances that craft an essential adjustment to the organization in whatever capacity is needed [15]. Thus, the use of this study indicates the demand for AI driven systems that will contribute to positive advances in organization, leading to better results in articulating safety demands for the systems.

Additionally, the use of AI enhances response time in contemporary environment of dynamic environment of threat landscapes. Continuous learning of the AI powered self-healing system through reinforcement learning creates a chance to handle new data and adapt to emerging threats. This maintains a high level of efficiency in every step of development on the threat landscape [16]. The adaptability of the systems creates an essential step in identifying and working towards the management of security with a rapid pace of change, marking an increased effectiveness for the organization as desired. Advancing the AI enabled system is therefore key to ensuring that threats can be identified and mitigated within the shortest time possible, creating a safe appeal for the company to handle its needed advances in securing every bit of advancement.

This study further stamps the instrumental contribution of human intervention to AI-driven decision-making appeals. The research indicates that operations of AI as a black box threaten transparency and trust, further leading to the use of a hybrid approach that retains human significance in decision making on interventions that can detail the growth of the organization to become a better and safer entity [17]. The organizational use of both human and AI will ensure both efficiency and speed in handling threats, as it leverages on AI to manage routine tasks and human interventions for critical decisions, leading to balance between automation and accountability inferences in the organization.

This research offers a great insight on the benefits of AI driven self-healing security systems through the reduction of cybersecurity workloads, improving real-time threat mitigation and enhancing the response efficiency of security systems in addressing dynamic threat landscapes of the modern organizations. The results indicate efficiency of the AI driven self-healing systems while retaining an incrementally beneficial scope of dealing with AI challenges such as decision making and accountability. Thus, companies have to focus on having a balance of AI-powered security systems and human experts to achieve a hybrid approach in sustaining and managing their needs at any given instance.

VI. CONCLUSION

A. Summary of Findings

This study indicates that AI-enabled self-healing security systems have a significantly better cyber resilience and can adapt based on their capabilities. The use of AI in handling anomaly detection and ensuring reinforcement learning creates an instrumental chance to have an autonomous detection, analysis and mitigation of cyber threats in real time. These AI enabled self-healing security mechanisms ensure that the organization is always secure and works towards achieving a remarkable scope of detailing and handling their requirements to a desired level. This study indicates that the use of the AI enabled self-healing security systems provide 96.8% detection rate, 75% improvement in system recovery time, 40% reduction in false positives, 60% reduction in downtimes and 35% reduction in false negatives. These results indicates that the use of the AI-powered self-healing security systems provide an enhanced security for organizations, providing a safeguard against sophistication and complexity of cyber threats.

B. Implications

These findings have an influence on organizations and Security Operations Centers (SOCs) that demand a stronger cybersecurity posture. The adoption of AI driven automation ensures that SOCs can have better threat detection and manage response capacities, reducing workload on human analysts, enabling them to focus on other strategic activities within the organization. The shift can ensure there is an increased operational efficiency which addresses shortage of skilled cybersecurity experts, making the use of AI incremental for the organization to leverage on cost savings as they accelerate incident response in any given instance. Thus, the potential of these self-learning systems to adapt to the evolving threats makes it much easier to ensure long term resilience for the organizations in the evolving threat landscape.

C. Future Research Directions

Researchers should focus on enhancing AI transparency by looking into ways to integrate Explainable AI (XAI) in the different elements of the security provision of the company. Nonetheless, the researchers have to look into AI and Human collaboration in self-healing security frameworks, to ensure optimal balance for human oversight, an element that will make organizations to have the most sufficient platform of engaging and ensuring their activities are well aligned and registered in the desired fashion as indicated in Table.1. Considerable advances in the capacity to handle adversarial threats is also a key research area, enabling organizations to achieve their demands in advancing and ensuring suitable development for the companies at any given instance. Thus, these trends have to be used to ensure further growth in security posture of organizations.

Area	Challenge	Solution
Decision Maki Transparency	Lack of explainability Decisions	Integrating Explainable AI
AI-Human Collaboration	Creating a balan between AI and Hum Activities	Developing Hybr Approach
Adversarial Threats	AI is open manipulation duri training	Adversarial a Robust Training

Table 1: Future Research Directions

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